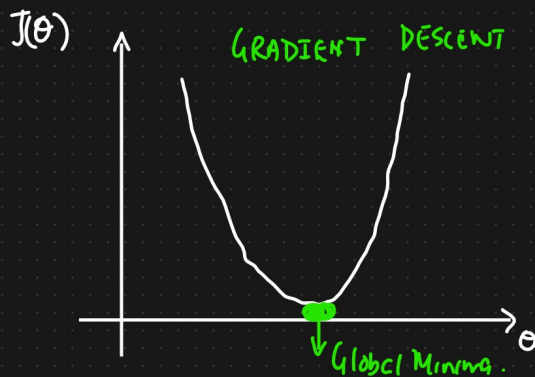
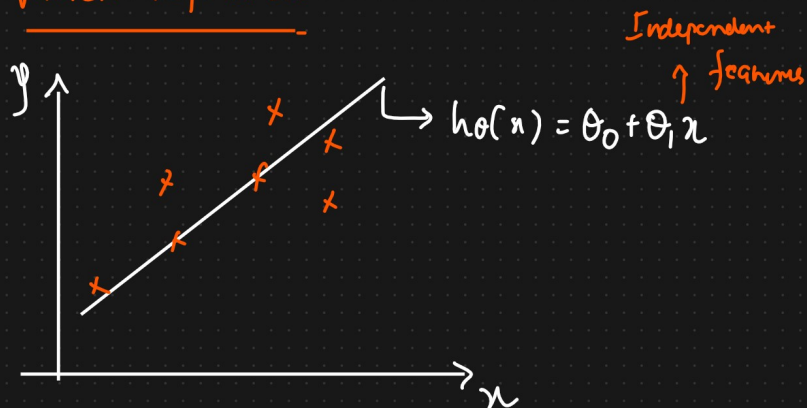


Ridge Regression, Lasso Regression, Elasticnet Regression

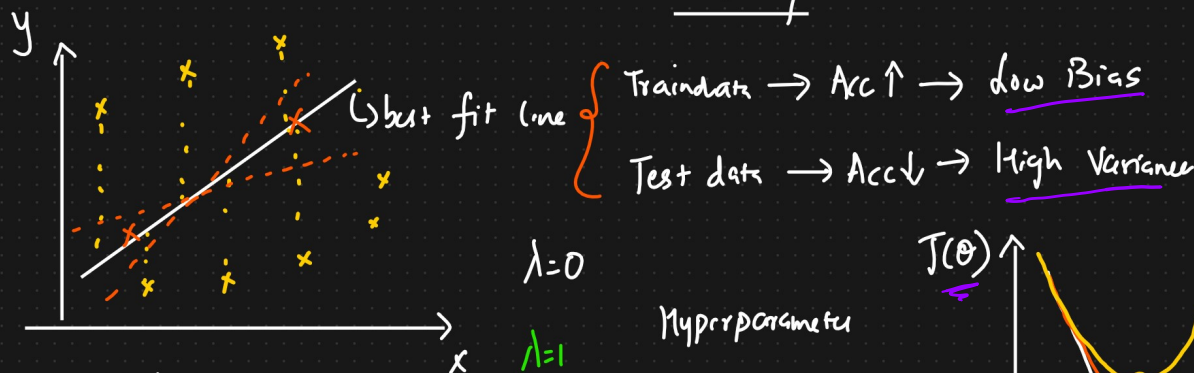
Linear Regression



$$\text{Cost fn} = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$$

Mean Squared Error

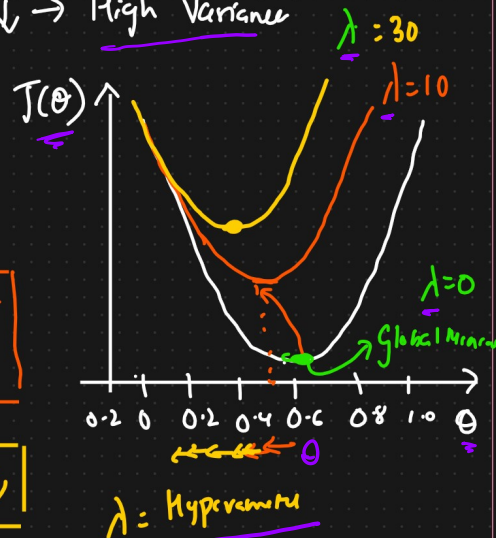
① Ridge Regression (L2 Regularization) → Reduce Overfitting [It working upon L2 norm]



$$\text{Cost fn} = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{i=1}^m (\text{slope})^2$$

Hyperparameter

$\lambda \uparrow$ slope ↓



$\lambda = 1$

> 0



$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

0.24

0.34 + 0.52x₁ + 0.48x₂ + 0.24x₃

After Applying Ridge regression.

5. After Applying Ridge Reg. value of θ decrease but not going to anytime "0" (zero).

$$= 0.34 + 0.40x_1 + 0.38x_2 + \boxed{0.14x_3}$$

② Lasso Regression (L_1 Regularization) $\rightarrow$ Feature Selection [It works upon L_1 norm]

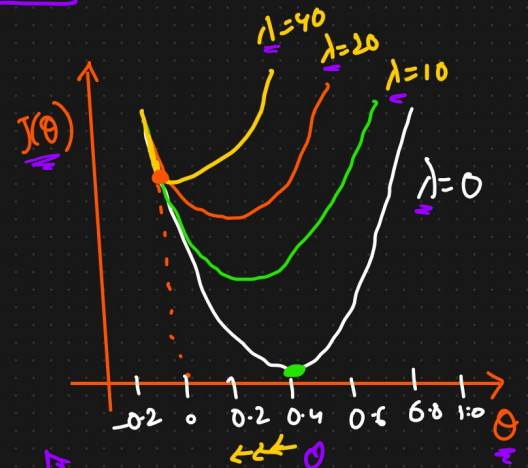
$$\text{Cost fn} = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{i=1}^n |\text{slope}|$$

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

$$h_{\theta}(x) = 0.52 + 0.65x_1 + 0.72x_2 + 0.34x_3 + \boxed{0.12x_4}$$

After Applying Lasso Regression

$$= 0.52 + 0.51x_1 + 0.60x_2 + 0.14x_3 + \boxed{0x_4}$$



At the end the best θ which value is very close to "0" (zero) is gone be here in Lasso Regression "0".

③ ElasticNet Regression

- $\rightarrow$ ① Reduce Overfitting
- $\rightarrow$ ② Feature Selection

$$\text{Cost fn} = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda_1 \sum_{i=1}^m (\text{slope})^2 + \lambda_2 \sum_{i=1}^m |\text{slope}|$$

It's called

Hyperparameter Tuning the Linear Regression

Reduce Overfitting

Feature Selection